

Integrated Mixture of Neighborhood and Community Experts for Graph-Based Fraud Detection

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Abstract

Graph-based fraud detection (GFD) aims to identify fraud nodes within graph-structured data that significantly deviate from the majority of benign nodes. However, existing graph neural networks (GNNs) often struggle in GFD scenarios due to their reliance on homophily assumption, which is frequently violated by the inherent homophily-heterophily mixture of fraud graphs. Moreover, most methods focus primarily on local topology, overlooking mesoscopic community structures, making them less efficient in detecting suspicious patterns like densely connected subgraphs. To address the aforementioned issues, we present NeCo, a novel approach that integrates mixture of neighborhood and community experts for graph-based fraud detection. Specifically, we first introduce a fraud-discriminative representation preservation mechanism from a neighborhood perspective, leveraging the empirical finding that fraud nodes tend to exhibit larger feature propagation discrepancies compared to benign nodes. We then design a community-oriented node representation module that models structural compactness among nodes, enabling the detection of suspicious topological patterns associated with fraud behaviors. By integrating these two complementary perspectives, NeCo can effectively capture both local inconsistency and global structural irregularity. Extensive experiments across five real-world datasets demonstrate the effectiveness of our proposed NeCo over state-of-the-art baselines.

CCS Concepts

• Computing methodologies → Machine learning algorithms.

Keywords

Graph Representation Learning, Graph Neural Network, Graph-Based Fraud Detection

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1 Introduction

With the rapid advancement in information technology, various types of fraud have surged dramatically in recent years, resulting in substantial economic losses worldwide. Graph-based fraud detection (GFD) [20, 26] has emerged as a powerful paradigm for identifying rare patterns that deviate from the majority patterns within graph-structured data, which has been extensively applied across diverse domains, such as financial transactions [6, 12], and cyber security [17]. At its core, GFD can be framed as a binary node classification task, whereby the goal is to classify a target node as either fraud or benign based on its features and topology.

Owing to the powerful capability in modeling graph-structured data, graph neural networks (GNNs) have gained significant traction as a promising approach for identifying frauds [14, 46]. Most typical GNNs built on the homophily assumption, wherein node representations are learned through iterative propagation and aggregation of information from neighboring nodes. However, fraud detection in real-world scenarios often involves a complex interplay of both homophily and heterophily. As shown in Figure 1, in some cases, fraud nodes frequently establish connections with benign nodes to evade detection (i.e., local inconsistency), while benign nodes predominantly interact with other benign nodes, reinforcing homophily. This hybrid interaction pattern deviates from the pure homophily assumption, posing challenges for GNNs to effectively identify and distinguish fraud behaviors. More critically, certain types of fraud involve collaboration among multiple nodes, leading to the formation of dense subgraphs or tightly connected clusters [32] (i.e., global structural irregularity). Such collaborative fraud further complicates the fraud detection task for existing GNNs, as they may struggle to differentiate between naturally occurring cohesive structures and those deliberately constructed by frauds,

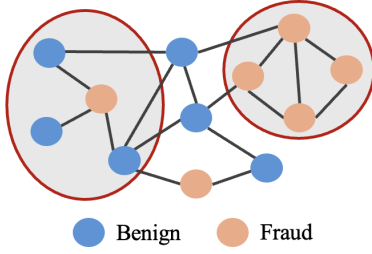


Figure 1: A toy example of fraud graph exhibiting both local inconsistency and global structural irregularity.

resulting in a higher risk of misclassifying fraud patterns as benign. Therefore, it is imperative to explore a new GNN approach capable of capturing both local inconsistency and global structural irregularity to enhance fraud detection.

Until then, a variety of tailored GNN-based fraud detection methods have been proposed to capture different types of fraud patterns, which can be broadly divided into two branches, including spectral-based and spatial-based methods. The spectral-based approaches focus on identifying frauds through spectral graph theory, where fraud behaviors are often linked to high-frequency components in a graph’s spectral representation. Typical examples include BWGNN [34] and GHRN [9]. However, these methods rely on expensive spectral decomposition that generally requires the entire graph input, resulting in high computational overhead, especially for large-scale graphs. Concurrently, the spatial-based approaches aim to design a new message passing mechanism that amplifies local prominence while enhancing global consistency. For example, GADAM [2] introduces an adaptive message passing with hybrid attention to allow each node selectively absorb fraud or benign signals from its surroundings. PMP [47] proposes a partitioning message passing paradigm by employing distinct aggregation functions to independently handle homophilic and heterophilic neighbors. However, these methods assume that all benign nodes exhibit uniform behavior, overlooking scenarios where benign nodes and their benign neighbors may span different clusters. This misalignment limits the ability of existing message passing mechanisms to effectively capture the diverse yet legitimate relational patterns, failing to detect subtle fraud patterns or even misclassify frauds as benigns.

To tackle the aforementioned issues, we propose a new GNN approach via multi-perspective modeling that integrates mixture of **N**eighborhood and **C**ommunity experts for fraud detection, termed NeCo. It contains two components: a neighborhood expert modeling and a community expert modeling. Specifically, from a neighborhood perspective, we explore what critical features should be preserved during information propagation for GFD. Based on these insights, we design a fraud-discriminative representation preservation mechanism that captures the discrepancy between an ego node and its surrounding neighborhood representations, enabling its outstanding ability to handle the coexistence of homophily-heterophily mixture. We then develop a community-oriented node representation learning module driven by random walk paths, incorporating an ego-community affinity loss that encourages nodes to align more closely with their corresponding community centers. In this way, the model enhances the distinction between naturally cohesive

communities and those deliberately engineered by frauds, effectively identifying manipulated community structures. The main contributions can be summarized as follows:

- We analyze the dilemma faced by existing GNN-based fraud detection methods in simultaneously capturing local inconsistency and global structural irregularity, both of which are essential for detecting sophisticated frauds.
- We propose NeCo, a novel GNN-based fraud detection approach that harnesses the complementary advantages of neighborhood and community experts to efficaciously capture diverse fraud patterns.
- Extensive experimental results on five public datasets demonstrate that our proposed NeCo achieves state-of-the-art performance compared with a range of baselines.

2 Preliminaries

Attributed Graph. Let $G = (\mathcal{V}, \mathcal{E}, X, Y)$ denote an attributed graph, where $\mathcal{V} = \{v_1, \dots, v_N\}$ and $\mathcal{E} = \{e_{ij}\} \subseteq V \times V$ are the sets of nodes and edges, respectively, X denotes a node feature matrix wherein the i -th row x_i is the feature vector of node v_i , and Y represents the label set of nodes. The topological structure of G is represented by an adjacency matrix $A = [a_{ij}] \in \{0, 1\}^{N \times N}$, where $a_{ij} = 1$ if node v_i connects to node v_j , otherwise $a_{ij} = 0$.

Clustering Coefficient. The clustering coefficient is a widely used measure for quantifying how clustered a node’s local neighborhood is, serving as the basis for many sophisticated structural motifs [37]. Mathematically, for a target node v_i , it can be defined as the proportion of triangles that are closed within the node’s 1-hop neighborhood as [10]:

$$C_i = \begin{cases} \frac{2 \cdot |\{e_{jk} \in \mathcal{E} \mid v_j, v_k \in \mathcal{N}_i\}|}{d_i(d_i - 1)}, & d_i \geq 2, \\ 0, & d_i \leq 1, \end{cases} \quad (1)$$

where $\mathcal{N}_i$ and d_i denote the neighborhood and degree of node v_i .

To further illustrate whether there is a difference in clustering coefficients among fraud and benign nodes in graph-based fraud detection, we conduct empirical analysis by taking four widely-used fraud graphs, i.e., Photo [24], Weibo [43], Amazon [23] and YelpChi [31] datasets as examples, and selecting the top 100 fraud nodes and 100 benign nodes with the highest degree. As shown in Figure 2, we can find that among high-degree nodes, fraud nodes exhibit significantly higher clustering coefficients than benign nodes on all four datasets, especially on Photo and Amazon datasets. This mainly because fraud nodes form densely interconnected subgroups, inflating their clustering coefficients, while normal high-degree nodes relatively show sparser neighbor connections, reflecting organic behavior [32]. Therefore, identifying frauds with structural irregularities from a clustering perspective may be a promising and feasible approach.

Graph-Based Fraud Detection. Given an attribute graph G , graph-based fraud detection (GFD) can be formulated as a node-level imbalanced binary classification problem. In most cases, frauds (the majority class) are viewed as positive with label 1, while benigns (the minority class) are considered as negative with label 0 [40]. This work focuses on semi-supervised GFD, where nodes of the whole graph are divided into the labeled ones Y_{train} and the unlabeled

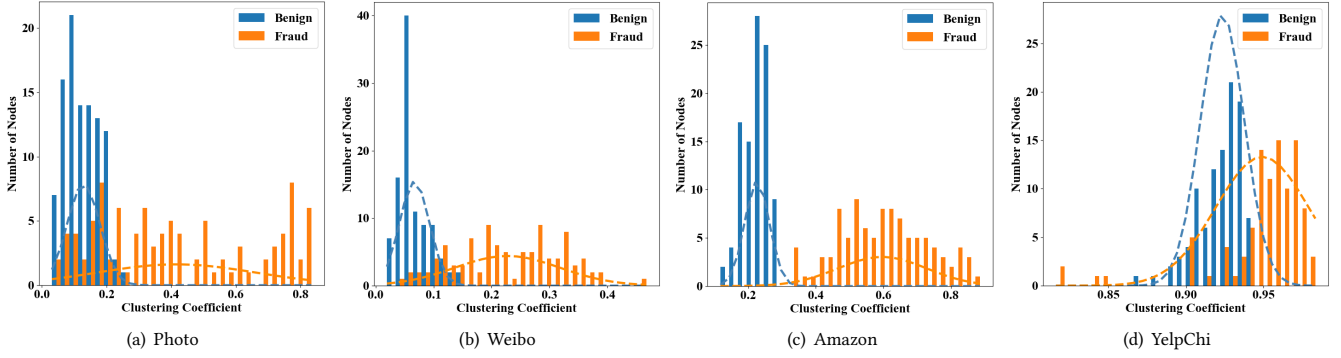


Figure 2: Histograms of clustering coefficients for the top 100 fraud nodes and 100 benign nodes with the highest degree on four graph-based fraud detection datasets.

ones Y_{test} not visible during training. Formally, the objective of GFD is to learn a mapping function $\mathcal{F}$ to calculate the unlabeled nodes with all known information, denoted as:

$$Y_{test} = \mathcal{F}(A, X, Y_{train}). \quad (2)$$

3 Methodology

We first briefly present the proposed method, and then introduce each component in detail.

3.1 Overview

To effectively tackle the inherent homophily-heterophily mixture in GFD scenarios while capturing both contextual and structural suspicious patterns, we propose a new approach for identifying frauds from the joint perspectives of neighborhood and community, namely NeCo. The whole architecture of NeCo is displayed in Figure 3, comprising two principal components, namely neighborhood expert modeling and community expert modeling. Specifically, we first introduce a neighborhood-aware residual representation mechanism that learns fraud-discriminative node representations by capturing feature propagation discrepancies. This is motivated by the observation that the discrepancy between the ego representation of the fraud node and its neighborhood representation tends to be larger than that of the benign node. We then design a community-oriented node representation module that identifies suspicious topological pattern based on the structural compactness of connections among nodes, as frauds often manifest as densely connected subgraphs. Finally, by integrating these two complementary perspectives, our proposed NeCo can effectively capture both local ego-neighborhood discrepancies and global ego-community irregularities, thereby enabling more robust fraud detection.

3.2 Neighborhood Expert Modeling

Considering that in GFD scenarios, fraud nodes typically submerged themselves among benign nodes through camouflage to reduce suspicion, showing heterophily, while the context surrounding benign nodes exhibits homophily. This makes it difficult for commonly used GNNs based on neighborhood smoothing to be effectively

generalized to GFD scenarios. In other words, fraud detection requires amplifying the difference between the target node and the neighborhood, while existing GNNs like GCN [14] and GraphSAGE [10] are dedicated to eliminating this difference, so that the representation of the fraud node is “assimilated” by the surrounding benign neighbors, resulting in significant attenuation of its fraud signal after multi-layer propagation.

Fraud-Discriminative Representation Preservation. Similar to [22], we consider capturing fraud signals as much as possible by analyzing discrepancy between the initial representation of the target node and its hidden representations. Specifically, we introduce SGC [39], a linearized variant of graph convolutional networks that iteratively smooths node features by propagating them through the graph topology without nonlinear activations, to aggregate node neighborhood information, and the k -th layer node representations can be formulated as:

$$H_N^{(k+1)} = \tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H_N^{(k)}, \quad Z_N^{(k+1)} = \text{MLP}(H_N^{(k+1)}), \quad (3)$$

where $H_N^{(0)}$ denotes the initial feature matrix, $\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}}$ ($\tilde{A} = A + I$) stands for the normalization of adjacency matrix with self-loops, $\tilde{D} = \text{diag}(\tilde{d}_1, \dots, \tilde{d}_N)$ and $\tilde{d}_i = \sum_j \tilde{a}_{ij}$.

After obtaining the hidden layer node representations, that is, $Z_N^{(1)}, \dots, Z_N^{(K)}$, the final node representations can be obtained by concatenating the residual between initial node representations and their hidden representations as:

$$R^{(k)} = Z_N^{(k)} - Z_N^{(0)}, \quad H_N = R^{(1)} \parallel \dots \parallel R^{(K)}, \quad (4)$$

where K denotes the number of layers, and $\parallel$ represents the concatenation function.

Ego-Neighborhood Affinity Loss. Inspired by [29], we introduce a neighborhood affinity loss, where the core idea is that the affinity between fraud nodes and their neighbors should be smaller than that of benign nodes. Mathematically, given a node v_i , the neighborhood affinity can be calculated by the similarity between representations of the target node and its neighbors as:

$$\gamma(v_i) = \frac{1}{|\mathcal{N}_i|} \sum_{v_j \in \mathcal{N}_i} \text{sim}(h_i, h_j). \quad (5)$$

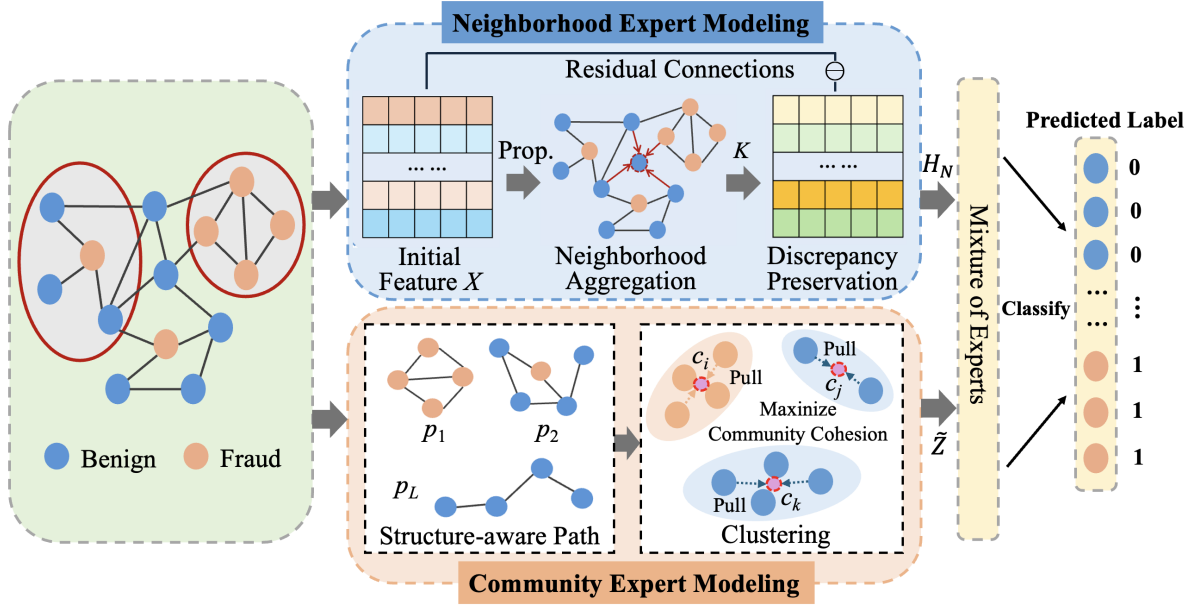


Figure 3: An overview of NeCo, which introduces multi-perspective modeling that integrates mixture of neighborhood and community experts for graph-based fraud detection.

Then, the neighborhood affinity loss can be formulated using a margin-based objective, which is asymmetric and enforces the separation of fraud and benign nodes based on their affinity as:

$$\mathcal{L}_N = \max(0, \tau - (\gamma(\mathcal{V}_f) - \gamma(\mathcal{V}_b))), \quad (6)$$

where $\gamma(\mathcal{V}_f) = \frac{1}{|\mathcal{V}_f|} \sum_{v_i \in \mathcal{V}_f} \gamma(v_i)$ and $\gamma(\mathcal{V}_b) = \frac{1}{|\mathcal{V}_b|} \sum_{v_i \in \mathcal{V}_b} \gamma(v_i)$ denote the average neighborhood affinity of the fraud and benign nodes, respectively, and τ is a hyperparameter used to control the margin of affinities between these two types of nodes. In this way, fraud nodes with local inconsistency can be inferred from a neighborhood perspective.

3.3 Community Expert Modeling

Considering that the neighborhood expert primarily focuses on capturing fraud signals within localized scopes, while some sophisticated GFD scenarios require coordinated efforts among multiple malicious entities, forming communities with distinctive topological patterns of “high cohesion, low coupling” [45]. That is, on the one hand, members of fraud community form closely connected subgraph structures through high-frequency transactions and intensive communications. On the other hand, the fraud community deliberately maintain the minimum necessary connection with the benign community to avoid detection.

Community-Oriented Node Representation Learning. Given that fraud nodes may exhibit higher clustering coefficients than benign nodes, we consider encoding structure information that intuitively characterize connectivity patterns of nodes for effectively identifying potential suspicious connectivity patterns in GFD scenarios from a community perspective. To this end, we employ restart random walks (RWR) [30], a principled sampling method that combine local structure probing through probabilistic restarts with

global structure awareness, thereby enabling revelation of both tightly-knit communities and their bridging connections across communities. Mathematically, let $P = \{p_1, p_2, \dots, p_L\}$ denote the random walk path set over all nodes with length L , the initial structure-aware node feature matrix can be obtained by adopting a path encoder f_{path} as:

$$H_S = f_{path}(p_1, p_2, \dots, p_L). \quad (7)$$

Here we choose Word2Vec [25] for path encoding due to its usability, which can be replaced by any other path encoder.

Then, in order to extract expressive low-dimensional node representations aiming at downstream fraud detection, we apply a multi-layer perceptron (MLP) to transform the initial structure-aware node features into soft labels as:

$$\tilde{Z} = \sigma(\text{MLP}(H_S)), \quad (8)$$

where $\tilde{Z}$ is the output of MLP, and σ denotes the activation function.

In light of the reliability of $\tilde{Z}$ plays a crucial role in subsequent community detection, we pre-train the MLP layer with the training ground-truth labels for several iteration [11]. The predicted probability that a given node v_i tending towards fraud can be calculated with a sigmoid operation as:

$$p_i = \text{sigmoid}(\tilde{z}_i). \quad (9)$$

Finally, considering the class imbalance phenomenon in GFD scenarios, the loss function is defined using weighted cross-entropy [44] as:

$$\mathcal{L}_{MLP} = - \sum_{v_i \in \mathcal{V}_L} [\delta y_i \log p_i + (1 - y_i) \log(1 - p_i)], \quad (10)$$

where δ is the ratio of fraud labels ($y_i = 1$) to benign labels ($y_i = 0$) in the training set.

With the obtained $\tilde{Z}$, we can infer the community information of each node using a specific clustering algorithm, and K -means is chosen here for identifying community.

Ego-Community Affinity Loss. To render node representations more distinguishable across different communities, drawing inspiration from [42], we adopt the Student's t -distribution [18], a heavy-tailed probability distribution particularly effective for measuring similarity in high-dimensional spaces, to quantify the representation similarity between node v_i and the j -th community center, represented as:

$$p_{ij} = \frac{(1 + \|\tilde{z}_i - c_j\|^2/\omega)^{-\frac{\omega+1}{2}}}{\sum_{j'} (1 + \|\tilde{z}_i - c_{j'}\|^2/\omega)^{-\frac{\omega+1}{2}}}, \quad (11)$$

where c_j is the j -th community center representation that is trainable, and ω is the degree of freedom of the Student's t -distribution. In this way, the probability that node v_i belongs to the j -th community can be obtained.

Then, to enforce nodes to align closer to their community centers (i.e., achieving high confidence community assignment), we enhance the community cohesion by normalizing the node-community similarities. The target similarity distribution can be calculated as:

$$q_{ij} = \frac{p_{ij}^2 / \sum_i p_{ij}}{\sum_{j'} (p_{ij'}^2 / \sum_i p_{ij'})}. \quad (12)$$

It is worth noting that the squared pairs are used to enlarge the similarities between similar instances while reducing that between different instances.

Finally, let $P = [p_{ij}] \in \mathbb{R}^{|\mathcal{V}| \times C}$ and $Q = [q_{ij}] \in \mathbb{R}^{|\mathcal{V}| \times C}$ represent the matrix forms of predict and target similarity distributions, respectively, the difference between these two distributions can be minimized by using Kullback-Leibler (KL) divergence [15] as:

$$\mathcal{L}_C = KL(Q \parallel P) = \frac{1}{|\mathcal{V}| \cdot C} \sum_{i=1}^{|\mathcal{V}|} \sum_{j=1}^C q_{ij} \log \frac{q_{ij}}{p_{ij}}, \quad (13)$$

where C denote the number of communities.

3.4 Model Optimization

In order to fully utilize neighborhood and community experts for identifying frauds, given a node v_i , we concatenate its representations h_i^N and $\tilde{z}_i$ learned from the perspectives of neighborhood and community, and employ a linear layer followed by a sigmoid function to calculate the predicted probability of the node tends towards fraud as:

$$\tilde{y}_i = \text{sigmoid} \left((h_i^N \parallel \tilde{z}_i) \cdot W \right), \quad (14)$$

where W is the trainable weight matrix, and $\parallel$ is the concatenation function.

Then, following the semi-supervised setting, we adopt weighted cross-entropy similar to MLP loss [44] as the loss function of fraud detection:

$$\mathcal{L}_{FD} = - \sum_{v_i \in \mathcal{V}_L} [\delta y_i \log \tilde{y}_i + (1 - y_i) \log(1 - \tilde{y}_i)]. \quad (15)$$

Finally, the overall loss function can be defined as:

$$\mathcal{L} = \mathcal{L}_{FD} + \alpha \mathcal{L}_{MLP} + \beta \mathcal{L}_N + \lambda \mathcal{L}_C, \quad (16)$$

where α , β and λ serve as hyperparameters that convey the weights of loss terms. In general, the detailed process of our proposed NeCo is summarized in Algorithm 1.

Algorithm 1 The proposed NeCo.

Input: An attributed graph $G = (\mathcal{V}, \mathcal{E}, X, Y)$, the number of training iterations T ;

Output: The classification result of each node in G .

- 1: **while** not reaching the number of iterations T **do**
 - 2: **for** each node v_i **do**
 - 3: Calculate node representations h_i^N by neighborhood expert using Eq. (5);
 - 4: Calculate node representations $\tilde{z}_i$ by neighborhood expert using Eq. (9);
 - 5: **end for**
 - 6: Fuse the node representations from neighborhood and community perspectives, and obtain the predicted label distribution with Eq. (15);
 - 7: Calculate the neighborhood affinity loss, MLP loss, ego-community affinity loss, and fraud detection loss via Eq. (7), (11), (14) and (16). Then obtain the final joint loss using Eq. (17);
 - 8: Back propagation and update trainable parameters;
 - 9: **end while**
-

4 Experiments

We begin by detailing the experimental setup, followed by a comprehensive evaluation of our NeCo against twelve state-of-the-art baselines. After that, we investigate how different training ratios influence the model performance, along with conducting an ablation study and parameter sensitivity.

4.1 Experimental Setup

Datasets	Photo	Weibo	Reddit	Amazon	YelpChi
#Nodes	7,535	8,405	10,984	11,944	45,954
#Edges	119,043	407,963	168,016	4,398,392	3,846,979
#Features	745	400	64	25	32
Fraud (%)	9.2	10.30	3.33	6.87	14.53

Table 1: Statistics of datasets.

Datasets. We conduct experiments on five real-world datasets related to GFD, where the key statistics of the datasets are presented in Table 1. Among them, Photo [24] is a co-purchase network constructed from Amazon, where nodes denote products and edges represents the co-purchase relationship among products. Weibo [43] is a user-hashtag graph sourced from the Tencent Weibo platform, where suspicious behaviors are defined as consecutive posts by a user within a specific temporal window. Reddit [16] represents a forum posts network derived from the social media platform Reddit, where banned users are labeled as frauds. Amazon [23] targets the detection of users who are compensated for writing fake reviews for products within the musical instrument category. YelpChi [31] focuses on identifying deceptive reviews on Yelp.com that are

Methods	Photo		Weibo		Reddit		Amazon		YelpChi	
	AUC	Macro-F1	AUC	Macro-F1	AUC	Macro-F1	AUC	Macro-F1	AUC	Macro-F1
GCN	97.87±0.53	89.75±0.64	96.29±0.19	92.75±0.38	60.76±0.44	52.84±0.74	79.53±1.74	61.86±1.78	54.28±1.44	53.25±1.06
GAT	96.79±0.52	88.56±1.28	94.46±1.10	88.41±3.32	65.72±0.54	51.09±0.59	78.09±10.53	72.95±10.54	53.20±1.36	52.50±0.70
GraphSage	77.61±17.02	69.75±16.87	87.00±0.71	88.77±0.61	66.57±1.10	52.69±0.67	94.69±1.37	90.27±0.92	77.11±0.23	64.62±0.07
AMNet	97.88±0.32	<u>91.00±0.21</u>	94.69±0.42	92.76±0.34	58.42±1.07	49.15±0.78	92.09±3.52	92.36±0.21	84.40±0.45	69.80±0.72
BWGNN	<u>98.26±0.37</u>	90.32±0.54	97.44±0.32	93.26±0.95	68.31±3.83	53.91±1.68	96.72±1.58	91.34±0.92	84.63±0.31	71.76±0.52
GHRN	96.81±0.12	87.35±1.21	96.17±0.23	91.01±0.36	65.88±1.20	53.00±0.64	95.50±0.92	92.38±0.37	83.51±0.44	70.46±0.48
SEC-GFD	97.27±0.29	88.24±0.49	<u>97.45±0.28</u>	<u>93.33±0.57</u>	<u>68.72±1.76</u>	<u>55.01±0.47</u>	97.31±0.10	<u>92.73±0.14</u>	<u>86.31±0.42</u>	<u>74.58±0.47</u>
CARE-GNN	90.62±0.63	77.28±0.91	91.95±0.60	79.08±1.00	59.25±8.09	41.42±3.84	95.55±0.12	87.66±0.26	70.75±3.01	60.89±1.67
PC-GNN	95.42±0.33	79.19±7.09	96.25±0.68	88.61±0.57	61.01±3.47	49.15±0.12	95.66±0.43	88.51±0.62	77.09±0.23	61.14±4.52
CONSIGAD	97.59±0.26	89.43±0.49	96.79±0.46	90.63±0.53	67.35±1.76	53.70±0.85	<u>97.63±0.11</u>	92.26±0.17	83.09±0.79	69.82±0.51
skl-GDN	96.95±0.04	90.84±0.08	96.27±0.30	91.34±0.55	67.57±0.72	49.16±0.48	96.02±0.13	90.49±0.21	76.30±0.12	62.33±0.39
ARC	74.40±1.14	39.60±5.80	88.80±0.28	33.52±6.89	57.49±3.11	35.92±2.22	79.43±2.55	39.80±7.53	61.73±0.63	42.46±1.72
NeCo	98.88±0.16	92.58±0.69	98.79±0.42	94.35±0.58	69.71±0.59	55.50±0.34	97.91±0.05	93.29±0.16	89.56±0.14	77.45±0.24

Table 2: Experiment results in terms of AUC (%) and Macro-F1 (%). The best and the second-best results are marked in bold and underline, respectively.

intended to artificially boost or undermine the reputation of certain products or businesses. It is worth noting that for both the Amazon and YelpChi datasets, we treat all types of edges as the same.

Baselines. We compare our proposed approach NeCo against three categories of twelve state-of-the-art approaches. They include 1) Generic GNNs: GCN [14], GAT [35] and GraphSage [10], which work under the homophily assumption; 2) Spectral GNNs for Fraud Detection: AMNet [1], BWGNN [34], GHRN [9], and SEC-GFD [40]; as well as 3) Spatial GNNs for Fraud Detection: CARE-GNN [4], PC-GNN [21], CONSIGAD [3], skl-GDN [8] and ARC [22].

Evaluation and Implementation. Following [7, 46], we employ AUC and Macro-F1 as evaluation metrics for GFD. To ensure fair comparison across all methods, we adopt the data splitting ratios of 4:2:4 for training, validation, and test set, with all experimental results reported as the average of 5 independent runs. For baselines, we implement GCN, GAT and GraphSage in DGL [36], while the remaining methods employ the source code released in the original paper. For NeCo, we adopt the Adam optimizer, where learning rate is 0.01, and weight decay rate is $1e-4$. We set the number of layers K in neighborhood expert modeling to 2. We tune the hyperparameters using grid search, with the length of restart random walk and the number of communities used in community expert modeling ranging from {5, 6, 7, 8, 9, 10} and {4, 8, 16, 32}, respectively.

4.2 Performance Comparison

To evaluate the effectiveness of our proposed NeCo, we conduct a comprehensive comparison with twelve baseline models on a fraud detection task. The experimental results with respect to AUC and Macro-F1 are presented in Table 2.

As shown, we can find that our proposed NeCo consistently achieves superior performance than all competitors across five datasets, confirming its effectiveness. Specifically, in terms of AUC, NeCo demonstrates performance improvements of 0.62%, 1.34%, 0.99%, 0.28%, and 3.25% than the best baseline method on Photo, Weibo, Reddit, Amazon, and YelpChi, respectively. In terms of Macro-F1, NeCo is 1.58%, 1.02%, 0.49%, 0.56%, and 2.87% more accurate than the best baseline method on these five datasets. These results highlight the superiority of preserving fraud-discriminative

representations by leveraging ego-neighborhood discrepancies to capture key fraud signals, while also underscoring the importance of detecting potential suspicious topological patterns from a community perspective. It is worth noting that our NeCo is significantly superior to the generalized GFD approach ARC, which identifies frauds based on in-context learning, especially on Photo and YelpChi datasets. This further validates the advantage of integrating mixture of neighborhood and community experts for effective fraud detection. Moreover, compared to classical GNN methods such as GCN, NeCo achieves notable gains of 1.01%, 2.50%, 8.95%, 18.38% and 35.28% in AUC, and 2.83%, 1.60%, 2.66%, 31.43% and 24.20% in Macro-F1 on Photo, Weibo, Reddit, Amazon, and YelpChi, respectively. The above performance gap highlights a fundamental limitation of GCN, which relies heavily on homophily assumption, hampering its ability to capture the homophily-heterophily mixed nature of fraud detection scenarios.

4.3 Analysis of Training Ratio

Given the limited availability of annotated data in real-world fraud detection scenarios, where manual labeling is often costly and time-consuming, we conduct experiments to assess the robustness and effectiveness of our proposed NeCo under different training ratios. Specifically, we compare NeCo with two SOTA baselines skl-GDN and SEC-GFD on Photo and Weibo datasets, respectively, with the ratios of training data vary from {5%, 10%, 20%, 40%}.

As presented in Figure 4, our proposed NeCo consistently outperforms the SOTA baselines skl-GDN and SEC-GFD across a range of training ratios in terms of Macro-F1, demonstrating its effectiveness even under limited supervision. This is primarily due to the fact that our NeCo simultaneously leverages both neighborhood and community experts to identify potential fraud signals from both microscopic and mesoscopic perspectives, thereby enabling a more holistic understanding of suspicious behaviors. Notably, on the Weibo dataset, our NeCo exhibits impressive stability compared to the other two baselines, showing only slight performance fluctuations across different training ratios, even at a low training ratio of 5%. These results further underscores the superiority and robustness of our model.

Methods	Photo		Weibo		Reddit		Amazon		YelpChi	
	AUC	Macro-F1	AUC	Macro-F1	AUC	Macro-F1	AUC	Macro-F1	AUC	Macro-F1
NeCo	98.88	92.58	98.79	94.35	69.71	55.50	97.91	93.29	89.56	77.45
- w/o Neighbor	96.48	86.75	96.55	92.01	66.74	56.09	97.34	90.89	84.21	71.58
- w/o Community	98.43	92.00	97.66	93.72	66.21	54.96	97.42	90.67	82.99	70.30
- w/o $\mathcal{L}_N$	97.86	87.92	97.02	94.05	68.51	55.22	97.65	91.20	71.76	61.46
NeCo (Sum)	98.03	91.92	97.91	94.08	63.13	53.61	95.47	88.26	85.99	73.45

Table 3: Comparisons of our NeCo with its four variants in terms of AUC (%) and Macro-F1 (%).

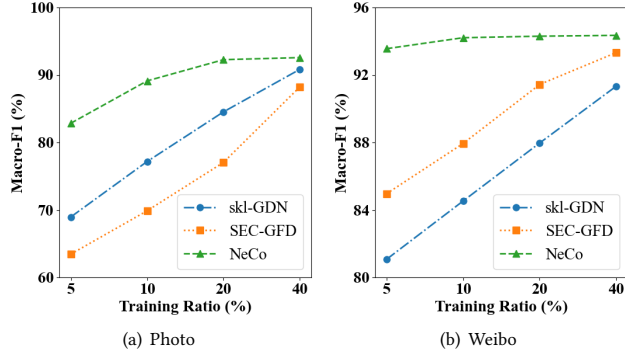


Figure 4: Performance comparison of NeCo, skl-GDN, and SEC-GFD across varying training ratios w.r.t. Macro-F1 (%).

4.4 Ablation Study

To further investigate the contribution of each component in NeCo, we perform ablation studies by removing each component at a time, that is, 1) NeCo of removing neighborhood expert modeling, named as w/o Neighbor; 2) NeCo of removing community expert modeling, named as w/o Community; 3) NeCo of removing ego-neighborhood affinity loss, named as w/o $\mathcal{L}_N$; as well as 4) NeCo of adopting summation operator instead of concatenation to integrate node representations learned from neighborhood and community perspectives, named as NeCo (Sum).

As shown in Table 3, we have the following observation: 1) NeCo consistently achieves superior performance compared to its four variants across all five datasets, highlighting the effectiveness of integrating both neighborhood and community expert perspectives for comprehensive fraud signal identification. 2) NeCo w/o Neighbor outperforms w/o Community on datasets like Reddit and YelpChi, where local node information plays a more critical role. In contrast, NeCo w/o Community achieves better performance on datasets like Photo and Weibo, where the detection of frauds tends to rely on community structures. These indicate that the contributions of neighborhood and community experts vary depending on the dataset characteristics. 3) After removing ego-neighborhood affinity loss, the performance of our proposed NeCo degrades across all five datasets, with particularly pronounced declines on the Amazon (and Yelp) dataset, where AUC and Macro-F1 drop by 2.44% (3.57%) and 5.03% (4.00%), respectively. This further validates highlight the critical role of the ego-neighborhood affinity loss in capturing local inconsistency among nodes, which contributes significantly to NeCo’s effectiveness across diverse datasets.

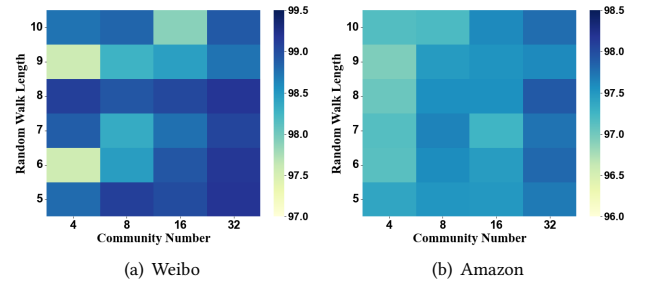


Figure 5: Sensitivity of random walk length L and community number C w.r.t. AUC (%).

4) Compared to NeCo (Sum), NeCo achieves an average improvement of 5.27% (and 4.18%) in AUC (Macro-F1) across five datasets, which illustrates the advantage of using concatenation operator over simple summation. This strategy preserves more nuanced and complementary information captured by different experts, thereby enhancing the model’s ability to detect frauds.

4.5 Parameter Sensitivity

Here we take the Weibo and Amazon datasets as examples to analyze the sensitivity of two key hyperparameters, namely the length of restart random walk L and the number of communities C used in community expert modeling, with AUC as the evaluation metric.

Analysis of Random Walk Length L . The random walk length L controls how far the restart-based exploration can propagate within the graph structure. We vary its value from 5 to 10 in steps of 1 and report the corresponding results in Figure 5(a). As observed, the model performance generally exhibits a trend of rising and then declining as the random walk length increases. Across Weibo and Amazon datasets, optimal performance is typically achieved when the random walk length is around 7 or 8. This is primarily because shorter walks may fail to capture sufficient structural information, limiting the model’s ability to recognize higher-order relational patterns. Conversely, too long walks may introduce noise, negatively affecting community partition. Thus, a moderate random walk length strikes a balance between capturing meaningful structural context and avoiding irrelevant information.

Analysis of Community Number C . The number of communities C determines the granularity of the mesoscopic structure, which in turn influences the model’s ability to detect suspicious patterns associated with frauds. We vary its value from 4 to 32, and present the corresponding results in Figure 5(b). As shown, the number

of communities has a significant influence on model performance. On both datasets, increasing the community number generally enhances performance up to a certain point. This improvement can be attributed to the fact that finer-grained community partition can better capture structural patterns, which are critical for distinguishing between benign and fraud behaviors. However, when the community number becomes too large, performance gains tend to plateau or even slightly deteriorate. This is likely due to over-segmentation, where meaningful structural cohesion is fragmented, undermining the effectiveness of community-oriented modeling.

5 Handling Camouflage Attributes

Considering that frauds in real-world may adjust their behaviors to camouflage their malicious activities, thereby undermining the effectiveness of models. Following [27], we evaluate the performance of NeCo in terms of AUC under situations where the raw attributes are camouflaged. Here we introduce camouflage by randomly replacing 10%, 20%, and 30% of the original attributes, assigning fraud nodes the mean feature values of benign nodes. As shown in Table 4, we can find that our proposed NeCo is robust against different levels of camouflaged feature, benefiting from its effective design of jointly leveraging the expert knowledge of neighborhood and community to uncover fraud behaviors.

Ratios	Photo	Weibo	Reddit	Amazon	YelpChi
0%	98.88	98.79	69.71	97.91	89.56
10%	98.16	98.56	69.54	96.38	86.44
20%	98.25	97.48	68.76	94.81	86.55
30%	97.60	96.70	68.14	95.18	86.36

Table 4: Performance under different ratios of camouflage attributes in terms of AUC (%).

6 Related Work

In recent years, Graph Neural Networks (GNNs) have attracted considerable attention for their ability to identify fraud nodes that exhibit significant deviations from the majority of benign ones. Existing GNN-based fraud detection can be broadly categorized into two main types, that is, spectral-based and spatial-based methods. More detailed surveys can be found in [28].

Spectral GNN for GFD. The spectral-based methods detect frauds by leveraging spectral graph theory, where the insight is that fraud behaviors are typically associated with high-frequency signals in graph’s spectral domain. For instance, AMNet [1] introduces an adaptive multi-frequency GNN for fraud detection that dynamically integrates low- and high-frequency signals. GHRN [9] introduces a prediction-aware edge indicator that evaluates post-aggregation similarity scores to prune potential heterophily edges. After that, SplitGNN [38] is a spectral GNN method that utilizes graph splitting and adaptive band-pass filtering to effectively detect frauds. SEC-GFD [40] investigates the association between heterophily edges and spectral energy distribution, accordingly designs a spectrum-enhanced and environment-constrained graph-based fraud detector. However, these approaches depend on costly spectral decomposition, which requires the entire graph input, leading to expensive computational overhead, particularly for large-scale graphs.

Spatial GNN for GFD. The Spatial-based methods typically detect fraud behavior through designing customized message passing mechanisms or employing data augmentation strategies. For example, H²-FDetector [33] designs a new GNN-based fraud detector that simultaneously consider both homophilic and heterophilic connections. AO-GNN [13] develops an AUC-oriented GNN approach that addresses the label imbalance from the standpoint of maximizing AUC metric. Thereafter, BSL [41] addresses the challenge of limited annotated labels in fraud detection scenarios by introducing a new barely supervised learning approach that leverages unlabeled samples as auxiliary signals during training. DGA-GNN [5] introduces a dynamic group aggregation strategy for tackling the issue of feature non-additivity and improving the discriminative of messages in fraud detection. Analogously, PMP [46] proposes a simple yet effective message passing paradigm that employs customized aggregation functions to separately integrate information from neighbors of different classes, so as to identify frauds. CGNN [19] introduces a context-aware GNN that leverages category semantic decomposition and denoising attention, while using feature augmentation and consistency regularization to enhance fraud detection under extremely limited labels. More recently, ARC [22] is a generalist GFD approach that identifies frauds across diverse graph datasets in an on-the-fly manner. However, many of these approaches presume consistency in benign node behaviors, failing to account for instances where benign nodes and their benign neighbors reside in distinct clusters. This mismatch reduces the effectiveness of message passing in modeling the legitimate diversity of relationships, ultimately causing subtle frauds to go undetected or be incorrectly labeled as benigns.

7 Conclusion

In this paper, we propose a new graph-based fraud detection approach NeCo that jointly leverages the complementary advantages of neighborhood and community experts, where the core idea is to effectively capture both local ego-neighborhood discrepancies and global ego-community irregularities. With the insight that fraud nodes typically exhibit larger divergence between their ego and neighborhood representations compared to benign nodes, we design a fraud-discriminative representation preservation mechanism to capture local inconsistency, thereby effectively mitigating the inherent homophily-heterophily mixture problem in fraud graphs. We then design a community-oriented representation learning module that captures the structural compactness among nodes, enabling the detection of suspicious fraud patterns by exploiting the “high cohesion, low coupling” properties inherent in communities. Extensive experiments demonstrate that our proposed NeCo achieves new state-of-the-art results on several public GFD datasets, underscoring the effectiveness of multi-perspective modeling in uncovering intricate fraud patterns.

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